



A Model for the airport gate assignment problem (AGAP) with greenhouse gas emissions considerations

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ARTICLE INFO

Keywords:

Airport gate assignment problem
Airports
Genetic algorithm
Nash equilibrium
GHG emissions

ABSTRACT

This paper investigates the problem of optimally assigning airplanes to airport gates (Airport Gate Assignment Problem—AGAP), with GHG emissions considerations. A multi-objective mathematical programming model is formulated with the objectives of minimizing flights assigned to apron gates, walking distances of connecting passengers, and GHG emissions because of airplane taxiing movements. A genetic algorithm, which incorporates a Nash equilibrium control element, is used to solve the problem. The Nash equilibrium element allows the fast and efficient identification of those genetic algorithm parameters that yield the best solution to the problem. The model and solution process are demonstrated in a hypothetical airport with ten gates, using simulated flight schedule data, which resemble operations of a real-world scaled-down airport with high passenger traffic. Results indicate considerable GHG emission savings and improved passenger walking burden at the airport over a simplified assignment approach. Specifically, the algorithm achieved a reduction of 1356.6 Kg CO₂ emissions compared to non-optimized flight distribution, i.e. an average reduction of about 19 liters per aircraft, as well as an additional indirect reduction in total CO₂ emissions due to a reduction in delays and congestion.

1. Introduction

The significant rise in air traffic over the years before the COVID-19 pandemic period has increased complexity in airport ground operations, which, along with other factors, causes delays in airplane departures (Zámková et al., 2019). The COVID-19 pandemic had a devastating impact on the airline industry, causing a significant decline in airport traffic (Sun et al., 2021; Wu et al., 2023). After the come back to stability, the rise continue and overload this period (Sun et al., 2023). Among them, air carrier attributed delays because of late airplane arrivals occur in about 63 % of delayed flights (Bureau of Transportation Statistics, 2025); these delays are usually attributed to inadequate airport operations and passenger traffic management (Eurocontrol, 2022, 2024). Areas of airports where delays occur include departure gates, taxiways, apron areas and runways, with taxiway-related procedures, runway usage, air and ground traffic congestion, weather conditions, and airplane malfunctions being major sources for them (Belobaba, 2009).

Airlines are also focusing on air service quality levels and passenger satisfaction. With delays, this is not possible because they often cause long queues at airports, long waiting times and therefore a not so

positive experience for passengers, as they can be a major source of negative emotions and anger for passengers (Kim & Park, 2016). Additionally, passengers need to be treated differently because of the difficulty in achieving relative passenger comfort and avoiding problems such as missed connections, baggage that did not reach the final destination and passenger inconvenience. For this reason their responses need to have three basic things: speed, efficiency and reliability (De Neufville & Odoni, 2013). Furthermore, air transportation is a GHG emission-intensive sector; even though air transportation is responsible for 2.5 % of global emissions (Lee et al., 2021), only 10 % of the world's population uses airplanes for travel (Gössling & Humpe, 2020). The outlook is not encouraging, since air transportation generated CO₂ emissions are expected to increase up to 22 % by 2050 (European Union (EU) 2015).

The air transportation industry is under growing pressure to reduce emissions as its increasing contribution to global greenhouse gases conflicts with international climate goals as the net-zero emissions target by 2050 (United Nations, 2021).

To achieve this, IATA estimates that it is essential to focus on more efficient and effective aircraft, using sustainable fuels (Wandelt et al., 2025), and implementing policies and strategies to reduce emissions

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<https://doi.org/10.1016/j.jatrs.2025.100074>

Received 19 March 2025; Received in revised form 12 May 2025; Accepted 13 May 2025

Available online 14 May 2025

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(Mello & Rosario, 2024). Ground delays do contribute to increased GHG emissions because of taxiing and idling with engines running while on the ground (Dissanayaka et al., 2020) and so the goal is to minimizing that as well.

Assigning airplanes to departure gates is a critical process in airport ground operations, as it affects passenger services and operations. The so-called “Airport Gate Assignment Problem (AGAP)” is that of allocating arriving airplanes to airport gates so that delays and operational costs are minimized while operational efficiency and a seamless passenger experience are achieved (Daş et al., 2020). The flight gate assignment problem is guided by decision-makers’ priorities, which generally fall into two categories: passenger-oriented and airport-oriented. Passenger-oriented factors focus on service quality and satisfaction, including passenger travel distance (Cheng et al., 2012; Ding et al., 2004, 2005; Haghani & Chen, 1998; Marinelli et al., 2015; Yan & Chang, 1998; Yan & Huo, 2001), terminal response time (Kim et al., 2013) baggage travel distance (Haghani & Chen, 1998) and passenger waiting times (Yan & Huo, 2001). However, airport-oriented factors emphasize efficiency and cost reduction, such as flight apron assignments (Ding et al., 2004, 2005; Marinelli et al. (2015).), flight-gate preferences (Dorndorf et al. (2008, 2012), and aircraft handling costs, including towing operations, gate reuse, and taxiing time (Dorndorf et al., 2008, 2012; Kim et al., 2013).

In this context, this paper proposes a model for the AGAP, which introduces the aspect of minimizing the carbon footprint of airplanes in the assignment process. Indeed, assigning airplanes to gates in such a way that aircraft waiting times, taxiing distances and transfer passengers walking distances are minimized may contribute to both improved operations and lower GHG emissions (Dissanayaka et al., 2020). The remainder of the paper is structured as follows: In Section 2, a review of the literature is offered in the subject, while Section 3 presents model developed for the study and the analysis of methodology. Section 4 showcases an application of the model in a hypothetical airport layout, and conclusions are provided in Section 5 of the paper together that propose further research.

2. Literature review

The focus of the airport gate assignment problem (AGAP) is to improve the allocation of airplanes to airport gates and essentially minimize associated passenger and operator costs. Different mathematical programming models have been introduced over the years for describing the AGAP, along with various techniques adopted for solving the associated mathematical programs (Bouras et al., 2014). In an early study, Mangoubi and Mathaisel (1985) applied a linear programming model and a heuristic for assigning airplanes to gates so that the total walking distances for passengers were minimized. Yan and Chang (1998) formulated the GAP as a network model, which considered different operational constraints and Haghani and Chen (1998) developed mathematical programming models for the AGAP, which optimized total walking distance for passengers and gate utilization. Multi-objective optimization models for gate assignment were proposed by Yan and Huo (2001), which considered passenger walking distances, gate utilization, and operational constraints. Dynamic assignment of gates to airplanes was tackled by Xu and Bailey (2001); their model was formulated as a mixed integer quadrating program, which attempted to minimize waiting times for both transferring and non-transferring passengers.

Ding et al. (2004, 2005) investigated the over-constrained AGAP and attempted to minimize gate conflicts and improve gate utilization of gate resources, using heuristics and applying a greedy algorithm and a neighborhood search method for minimizing unassigned flights to a gate and passenger walking distance. Cheng et al. (2012) explored the application of meta-heuristics to the airport gate assignment problem (AGAP) and concluded that meta-heuristics can produce efficient solutions within a reasonable timeframe and Kim et al. (2013) developed a

comprehensive scheduling framework that integrates various aspects of airport operations to optimize gate assignments. Minimizing passenger transit time and gate idle time was the focus of Marinelli et al. (2015).

In a series of papers, by Dondorf et al. (2008, 2012) the AGAP was modeled as a clique partitioning problem with an or on maintaining robust schedules concerning a reference schedule. Maharjan and Matis (2011) converted the AGAP into a multi-commodity flow network model, which minimized total walking distance for passengers during gate reassignment, and Kumar and Bierlaire (2014) modeled the AGAP by concentrating on connection revenues, minimizing zone usage costs, and increasing the robustness of gate assignments. An alternative approach based on quantum computers and quadratic unconstrained binary optimization for modeling the AGAP was proposed by Stollenwerk et al. (2018). A mixed integer linear programming model was developed by Silva et al. (2020) for the GAP, which considered maximization of airport revenues and passenger behavior. Similarly, Wen and He (2020) focused on passenger satisfaction during layovers for the AGAP by minimizing waiting times in their model.

Asadi et al. (2021) developed a hybrid meta-heuristic algorithm by combining the Shuffled Frog-Leaping Algorithm (SFLA) and the Grasshopper Optimization Algorithm (GOA). This approach overcame the drawbacks of each algorithm, achieving an optimal solution with reduced computational costs, and was tested using a scenario from Frankfurt Airport. Bi et al. (2022) proposed a Branch-and-Price algorithm (BAPA) solution and improved the solution speed by adding a heuristic search with a certain probability in each search node and verifying the model's performance by using actual flight data of Kunming Changshui International Airport, China. Karsu and Solyali (2022) developed a mixed-integer linear programming model to minimize apron usage and passenger travel distances within the airport. More recently, Shan et al. (2023) indirectly incorporated GHG emission considerations in their AGAP model, aiming to optimize flight assignments, fuel consumption, and gate assignment robustness. Das and Gzara (2024) presented a column-generation-based algorithm to generate gate schedules that minimized squared slack time while reducing passenger walking distances. Recently, Cao et al. (2024) proposed a more recent approach, introducing a predict-then-optimize framework; a CNN-LSTM-Attention model was used to predict flight arrival times and employed a bi-objective optimization model for scheduling. Published papers are summarized in Table 1:

The concept of Nash Equilibrium offers a realistic and widely accepted solution framework for strategic games where there are no strictly dominant strategies. In a Nash Equilibrium, each player chooses a strategy that is an optimal response to the strategies of the others, with no incentive to unilaterally change it (Nash, 1950). This equilibrium requires rational behavior, shared knowledge of rationality, and aligned expectations among players, ensuring stability of the system under individual deviations (Osborne, 2002).

Overall, the rich literature on the AGAP largely focuses on maximizing passenger comfort, by minimizing walking distances traveled by passengers and layover times. Some studies focus on optimizing airport revenues, by minimizing luggage transportation distances, vacant gates, and the time that these remain without an assigned airplane, flight gate preferences, taxiing distances, and airplane handling costs. However, only a handful of studies indirectly consider the environmental impacts of ground operations related to assigning arriving airplanes to gates. In this context, this paper contributes to the literature by proposing and demonstrating a new model and a meta-heuristic model for representing and solving the AGAP. The model considers both passenger aspects walking distances for connecting flights) and GHG emissions produced because of airplane ground operations (taxiing).

Table 1

Publications on the gate assignment problem.

Title	Authors	Year	Target	Algorithm
Predicting flight arrival times with deep learning: A strategy for minimizing potential conflicts in gate assignment	Cao et al.	2024	Maximize proportion of flights assigned to jet bridge gates among all flights, minimize total arrival taxiing distance	A CNN-LSTM-Attention model to predict flight, deep learning, branch-and-price algorithm, predict-then-optimize framework
Multiobjective Gate Assignment Model Considering Carbon Emissions	Shan et al.	2023	Reduce carbon emissions through optimal gate assignment	Non-dominated Sorting Genetic Algorithm-II (NSGA-II)
A New Formulation and an Effective Matheuristic for the Airport Gate Assignment Problem	Karsu and Solyali	2022	Minimize apron usage and total passenger walking distance	Matheuristic, Mixed integer linear programming (MILP)
The airport gate assignment problem: A Branch-and-Price Approach for improving utilization of jetways	Bi et al.	2022	Maximize the number of passengers using jetways	Branch- and-Price algorithm (BAPA)
Optimal schedule recovery for the aircraft gate assignment with constrained resources	Asadi et al.	2021	Minimize the total cost caused by delay occurred	Hybrid meta-heuristic algorithm by combining the Shuffled Frog-Leaping Algorithm (SFLA) and the Grasshopper Optimization Algorithm (GOA).
Airport Gate Assignment Problem considering Connecting Passengers	Wen and He	2020	Improve gate utilization, reduce connection time for connecting passengers	Genetic algorithm with data preprocessing (DP-GA)
Gate Assignment Problem: a case study of Lisbon airport	Silva et al.	2020	Combine the gate assignment problem with passenger behavior modeling, maximize airport revenues	Mixed-integer linear programming model (MILP), Discrete choice modelling (DCM)
Flight Gate Assignment with a Quantum Annealer	Stollenwerk et al.	2018	Minimize transit time for passengers	D-Wave Quantum annealing
A Metaheuristic Approach to Solve the Flight Gate Assignment Problem	Marinelli et al.	2015	Minimize passenger transit time and gate idle time	Bee Colony Optimization (BCO)

Table 1 (continued)

Title	Authors	Year	Target	Algorithm
Multi-objective Airport Gate Assignment Problem in Planning and Operations	Prem Kumar and Bierlaire	2014	Minimize zone usage costs, enhance gate plan robustness, and recover schedules during disruptions	Optimization programming language
Airport Gate Scheduling for Passengers, Aircraft, and Operations	Kim et al.	2013	Improve passenger experience while ensuring robust air-side operations	Balancing objective function, Tabu search
The use of meta-heuristics for airport gate assignment	Cheng et al.	2012	Minimize total passenger walking distance	Genetic algorithm, Tabu search, Simulated annealing
An Optimization Model for Gate Reassignment in Response to Flight Delays	Maharjan and Matis	2011	Minimize total walking distance for passengers during gate reassignment	Binary integer programming
Flight Gate Scheduling with Respect to a Reference Schedule	Dorndorf et al.	2012	Maintain robust schedules concerning	Clique partitioning problem, Heuristic algorithm
Modelling Robust Flight-Gate Scheduling as a Clique Partitioning Problem	Dorndorf et al.	2008	Maximize robustness and preference scores while minimizing unassigned flights and tows	Heuristic based on the ejection chain algorithm
The Over-Constrained Airport Gate Assignment Problem	Ding et al.	2005	Minimize ungated flights and passenger connection distances	Simulated annealing, Hybrid tabu search
Aircraft and Gate Scheduling Optimization at Airports	Ding et al.	2004	Minimize ungated flights and total passenger walking distance	Greedy algorithm
The Airport Gate Assignment Problem: Mathematical Model and a Tabu Search Algorithm	Xu and Bailey	2001	Minimize passenger connection times	Mixed-integer quadrating program
Optimization of multiple objective gate assignments	Yan and Huo	2001	Minimize passenger walking distances, maximize gate utilization	Multiple objective zero/one integer program, weighting method, column generation approach, simplex, branch-and-bound technique
Optimizing Gate Assignments at Airport Terminals	Haghani and Chen	1998	Minimize passenger walking distance and gate utilization	Heuristic, Integer programming
A Network Model for Gate Assignment	Yan and Chang	1998	Minimize passenger walking	Multi-commodity network flow,

(continued on next page)

Table 1 (continued)

Title	Authors	Year	Target	Algorithm
Optimizing Gate Assignments at Airport Terminals	Mangoubi and Mathaisel	1985	distance and cost Minimize total walking distances for passengers	Lagrangian relaxation (1) a linear programming relaxation of an integer program formulation and (2) a heuristic

3. Methodology

3.1. Model

Without loss of generality, the proposed model considers two planning objectives: passenger comfort and environmentally sustainable operations. The former is passenger-oriented and addressed by minimizing the number of airplanes assigned to an apron gate and the walking distance of travelers in connecting flights (Dorndorf et al., 2007). Indeed, walking and/or using a bus to reach an apron gate, as well as walking to a connecting flight are factors strongly affecting the comfort of air travelers and it affects the possibility of being late for the next flight when the transferring time is short. The latter planning objective is introduced in this study, by attempting to minimize GHG produced during taxing. In this context, let:

G^T	Set of Gates in the airport terminal 1,2,...j,...,m) where j, m random gates;
G^A	Set of Gates in the apron area of the airport 1,2,...n,...,o) where n,o random gates;
F	Set of Flights taking off/landing at airport 1,2,...i,...,k) where i,k random flights;
g	Total number of gates;
f	Total number of flights;
a_i	Arrival time of flight i at the gate, $\forall i \in F$;
b_i	Departure time of flight i from the gate $\forall i \in F$;
$wd_{k,l}$	Distance between gates for connecting passengers;
$GHG_{j,rs}$	GHG emissions produced between gate and runway take-off point;
$GHG_{Rf,j}$	GHG emissions produced between runway landing point and gate;
$p_{i,k}$	Number of connecting passengers between flight i and flight k $\forall i, \forall k \in F$;
s_i	Aircraft Size Category (III, IV, V);
s_g	Gate Size Category (1,2,3);
T_b	Minimum time interval the gate must remain empty between two successive airplanes;
T_{ta}	Minimum time the aircraft stays at the gate for necessary procedures;
w_i	Objective function weights;
$y_{i,j}$	Binary variable (1 if gate j can accept a plane type, 0 otherwise);
$x_{i,j}$	Binary decision variable (1 if flight i is assigned to gate j , 0 otherwise) $\forall i \in G, j \in F$;
$z_{i,j}$	Binary variable (1 if gate j accepts international flights, 0 otherwise).

The model formulation is shown below:

$$\min z = w_1 \cdot \sum_{n \in G^A} \sum_{i \in F} x_{i,j} + w_2 \cdot \sum_{j \in G^T} \sum_{m \in G^T} \sum_{k \in F} \sum_{i \in F} x_{i,j} \cdot x_{k,m} \cdot p_{i,k} \cdot wd_{j,m} + w_3 \cdot \sum_{j \in G^T} \sum_{i \in F} x_{i,j} \cdot (GHG_{j,rs} + GHG_{Rf,j}) \quad (1)$$

subject to:

$$\sum_{j \in G} x_{ij} = 1 \forall i \in F \quad (2)$$

$$\sum_{j \in G} x_{ij} \leq 1 \forall i \in F \quad (3)$$

$$|b_i - a_i| \geq T_{ta} \text{ for } x_{ij} = 1 \text{ and } x_{kj} = 1, \forall i \neq k \in F \quad (4)$$

$$x_{ij} + x_{kj} \leq 1 \forall j \in G \quad (5)$$

$$|a_i - b_k| \geq T_b \text{ for } x_{ij} = 1 \text{ and } x_{kj} = 1, \forall i \neq k \in F \quad (6)$$

$$x_{ij} \in \{0, 1\}, \forall i \in F, j \in G; y_{ij} \in \{0, 1\}, \forall i \in F, j \in G; z_{ij} \in \{0, 1\}, \forall i \in F, j \in G \quad (7)$$

The objective function (1) consists of three weighted components: the first component minimizes the total number of flights corresponding to gates at the apron. There are benefits to having flights loaded at gates in the terminal building instead of the apron: fewer flights in the apron area increase efficiency and facilitate passenger boarding, and there are no emissions from airport buses. The second component reflects the walking distance of travelers from their arrival to their connecting flight. The third component minimizes the total GHG emissions of airplanes moving from their landing spots to their assigned gates and from their assigned gates to the take-off positions for their next flight as a function of airplane consumption, taxiing distance, and idling time. Constraints (2) assign one gate per airplane simultaneously, and constraints (3) ensure that an aircraft can only be assigned to a gate and not be reassigned to a second gate. Constraints (4) set the minimum turnaround time, and constraints (5) ensure that airplanes whose turnaround timings overlap cannot be assigned to the same gate. Constraints (6) set the minimum time between successive airplanes assigned to a gate. Finally, constraints (7) are gate type and size constraints, which dictate restrictions on the sizes of airplanes that can be accommodated by a gate and whether a gate can accommodate flights from specific destinations (domestic, international).

2.2. Solution algorithm

A genetic algorithm (GA) is used for solving the model; GAs are selected because they offer a large solution space exploration, making it ideal for navigating complex and extensive search spaces. Assignment problems, such as AGAP, can be complex problems because of the large number of gates and flights. Also, the algorithm is well-suited for multi-objective optimization, enabling the simultaneous optimization of multiple criteria, such as taxiing distance and passenger comfort. Indeed, according to the literature review on the AGAP, several studies have exploited GAs for tackling the problem at hand. The proposed GA incorporates a Nash equilibrium control, which aids in both calibrating GA parameters and act as a local optimization mechanism. The rationale behind the Nash equilibrium in a GA is that of ensuring that no single parameter can unilaterally change its value to improve the overall GA performance while other parameters remain fixed.

The general structure of the GA with Nash control is straightforward: the algorithm starts by setting the GA parameters (number of generations, population size, etc.) and the problem data related to flights and gates. An initial population of candidate solutions is generated (population initialization); to ensure these starting solutions are feasible, a solution repair process is performed concerning gate type and size compatibility, as not all aircraft are accepted at all gates, and time overlaps cannot occur. For that initial population of solutions, the objective function (1) is calculated, which acts as the GA candidate solution fitness measure. The Genetic Algorithm (GA) selects parents based on fitness and performs crossover to exchange genetic information, creating diverse offspring. Mutation is applied to introduce random changes, maintaining variety and preventing the algorithm from

converging prematurely. To further avoid local minima, elitism is incorporated, preserving the top solutions each generation. Additionally, strategies like diverse selection methods and higher mutation rates help the GA explore new solutions, ultimately improving global optimization by balancing exploration and exploitation. The Nash equilibrium control is then applied: for each flight, its possible assignment to every available gate (including space on the runway) is tested. At each trial, the fitness of the modified solution is recalculated. If there is even one case where changing the gate of a single flight leads to a better fitness function value, then the solution is not considered a Nash Equilibrium. Conversely, if no gate change for any flight results in an improvement, the solution is regarded as being in Nash Equilibrium. Essentially, this process performs a local optimization check, assessing whether the solution is stable against individual changes, without requiring simultaneous adjustments for all flights. The application of the Nash control allows therefore to efficiently evaluate the performance of different GA parameters for the same number of generations, and also perform a local improvement of GA solutions for that number of generations. In terms of the gate assignment problem each flight is treated as a "player" making decisions about which gate to allocate, with the allocation influenced by the actions of other flights (i.e., other players). The Nash Equilibrium (NE) concept is applied as a form of local optimization, where every small modification to the solution is evaluated. This ensures that the final solution cannot be easily improved through unilateral changes by any single flight. While the concept of Nash Equilibrium is referenced, the problem is not formally defined or solved as a game theory problem. Consequently, traditional game-theoretic mathematical formulations (e.g., payoff calculation) are not employed. The genetic algorithm ensures solution stability by verifying that no individual flight repositioning can improve the overall fitness of the solution. If a solution is in NE, no single flight can change its assignment without negatively impacting the overall system performance and fitness. The genetic algorithm (GA) terminates after a maximum number of generations or convergence of solutions is reached according to the Nash equilibrium control. Note that GHG emissions in the fitness function are calculated according to the fuel consumption of each aircraft while on the ground and the total taxi distance traveled, assuming an average taxiing speed and using appropriate conversion factors.

4. Application

4.1. Overview and data

To validate our model, we apply it to a hypothetical airport layout that accommodates domestic and international flights to test our methodology under controlled conditions. Our example airport is based on a scaled-down version of a real-world, high-passenger airport. That means that simulate the ratio of aircraft to gates of an airport with high airport traffic volume because we want to test out the model in difficult situations. These figures are comparable to Gatwick Airport, UK South Terminal, with a ratio of 1.71 (London Gatwick Airport, 2025) and Newark Airport in New Jersey, whose Terminal A has a ratio of 2.03 (Aviation Newark Liberty International Airport, 2025). For instance, Newark Liberty International Airport (EWR) in a typical four-hour time slot in one terminal, approximately 60 aircraft require gate assignments across 30 gates. To create a manageable yet representative case, we proportionally scale this scenario by a factor of three, resulting in an airport with 20 aircraft and 10 gates. This approach ensures that the complexity of the scheduling problem remains realistic while allowing for efficient testing and analysis. The airport has a single terminal with ten contact gates, an apron area, and one runway. A simulated, four-hour daily peak schedule is set up, with incoming flights assumed to arrive every 10–20 min during peaks. This enables us to capture realistic scheduling dynamics without excessive computational overhead. Flights are categorized as either domestic or international and arrive at the same terminal, with only some gates being allowed to handle

international flights and while all flights use the same terminal, only a subset of gates is designated for international arrivals due to customs and security requirements. The airport accommodates three common aircraft design groups (III, IV, and V) based on FAA classifications (Federal Aviation Administration, 2024). These categories account for differences in aircraft length and wingspan. For reference, an Airbus A320 falls into Category III, while a Boeing 777 belongs to Category V. The fuel consumption estimation based on the basic fuel consumption value with further adjustment to the size of the aircrafts.

The model can be extended to incorporate additional complexities, such as longer connection distances in multi-terminal airports and enhanced security screening procedures for international arrivals. Following Table 2, Table 3 and Table 4 present the data used in this application.

4.2. Results

GA parameters should be calibrated so that those with the best performance are selected. Therefore, three population sizes (40, 50, and 60), three crossover rates (0.2, 0.4, and 0.6), and three mutation rates (0.05, 0.1, and 0.15) are selected. In addition, weights set for the objective function are set without loss of generality to $w_1 = 0.3$, $w_2 = 0.2$, and $w_3 = 0.5$, prioritizing reducing GHG emissions in airport operations. The GA is executed 20 times for each parameter combination. GA calibration results are presented in Table 5. If a solution reaches Nash equilibrium according to the GA control, then no reassignment of gates will lead to a better solution. As per the results, the most critical parameter is the crossover rate because the function value of 0.2 provides the best outcomes for all other combinations and a positive Nash equilibrium. The execution time of the algorithm for 100 generations is approximately 19 s.

According to the calibration results, it is evident that the objective function exhibits variations across different parameters and multiple runs. The lowest value observed is $-37,000.00$, while the highest value is $-39,875.00$. In all instances where the program indicated that Nash equilibrium was "FALSE," (meaning no equilibrium), this occurred from 20 iterations, and none resulted in a lower outcome. This suggests that although there might have been a better solution, the parameters at each instance failed to detect it. When a solution yields a positive Nash equilibrium "TRUE", one of these solutions is approached the optimal solution. Additionally, solutions with positive Nash equilibrium were re-examined to explore if any iteration provided better results. While GA is a heuristic method and thus cannot guarantee optimality, with this approach we converge toward a highly efficient solution.

To compare two different assignments before and after the GA application, a simple first-fit assignment was adopted: airplanes are assigned to gates sequentially based on ID, considering type and size constraints. That is, airplanes are placed at the first available gate in ascending order) that meets these constraints. GHG emissions are then calculated for this non-optimized assignment, which only considers constraints and selects gates based on the gate's ID. If none of the gates

Table 2
Airport gate data.

Gate ID	Destination	Gate Size	Taxiing Distance from Landing point (m)	Taxiing Distance to take off point (m)
A1	International	Size 3	1500	2600
A2	International	Size 2	1400	2400
A3	International	Size 3	1200	1800
A4	Domestic	Size 1	2000	2900
A5	Domestic	Size 3	1300	1900
A6	Domestic	Size 2	1100	1700
A7	International	Size 3	1600	2700
A8	International	Size 3	1450	2500
A9	Domestic	Size 2	1250	1850
A10	Domestic	Size 3	1950	2850

Table 3
Walking distance (m) between gates.

Gate ID	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	Apron
A1	0	75	150	225	300	375	450	525	600	675	2500
A2	75	0	75	150	225	300	375	450	525	600	2400
A3	150	75	0	75	150	225	300	375	450	525	2300
A4	225	150	75	0	75	150	225	300	375	450	2200
A5	300	225	150	75	0	75	150	225	300	375	2100
A6	375	300	225	150	75	0	75	150	225	300	2000
A7	450	375	300	225	150	75	0	75	150	225	1900
A8	525	450	375	300	225	150	75	0	75	150	1800
A9	600	525	450	375	300	225	150	75	0	75	1700
A10	675	600	525	450	375	300	225	150	75	0	1600
Apron	2500	2400	2300	2200	2100	2000	1900	1800	1700	1600	0

Table 4
Flight data.

Flight ID	Arrival Time	Departure Time	Aircraft Class	Destination	Connection	Fuel Consumption (kg/s)
AF1234	08:00	08:45	IV	International	DY9012	0248
BA2345	08:15	09:25	III	Domestic		0182
LH3456	08:30	09:40	V	International		0218
KL4567	08:40	09:30	IV	Domestic		0218
AZ5678	09:00	10:15	IV	International		0,15
IB6789	09:10	10:00	IV	International		0456
SK7890	09:30	10:30	III	Domestic		0208
CB8901	09:45	11:00	V	International		0456
ML9801	09:55	11:10	IV	International		0208
DY9012	10:00	10:30	IV	Domestic	AF1235	0204
MY9012	10:15	10:45	V	International		0456
KB6789	10:30	11:10	IV	Domestic	BA9120	0204
TT8890	10:45	11:40	IV	Domestic		0208
PT8890	10:55	12:00	IV	Domestic		0208
XA8910	11:00	12:10	V	International		0540
BA9120	11:15	12:15	III	Domestic	KB6789	0182
SK0236	11:30	12:30	IV	Domestic		0204
TK0123	11:35	12:55	IV	Domestic		0218
AF2345	11:45	12:45	III	Domestic		0182
TK7890	12:00	14:00	IV	International		0204

Table 5
Calibration results.

ID	Population	Crossover Rate	Mutation Rate	Nash Equilibrium	Best Fitness Measure Value	GHG emissions Reduction (Kg)
1	30	0.2	0.05	TRUE	−37,700.0	1341.288
2	30	0.2	0.10	TRUE	−37,000.0	1344.580
3	30	0.2	0.15	TRUE	−37,350.0	540.623
4	30	0.4	0.05	FALSE	−38,350.0	962.154
5	30	0.4	0.10	FALSE	−38,550.0	558.596
6	30	0.4	0.15	FALSE	−38,750.0	269.522
7	30	0.6	0.05	FALSE	−38,675.0	888.1575
8	30	0.6	0.10	FALSE	−39,125.0	617.649
9	30	0.6	0.15	FALSE	−38,500.0	594.146
10	40	0.2	0.05	TRUE	−37,875.0	915.676
11	40	0.2	0.10	TRUE	−37,275.0	649.907
12	40	0.2	0.15	TRUE	−37,000.0	1356.562
13	40	0.4	0.05	FALSE	−38,425.0	310.075
14	40	0.4	0.10	FALSE	−38,750.0	705.931
15	40	0.4	0.15	FALSE	−38,750.0	1094.216
16	40	0.6	0.05	FALSE	−39,375.0	382.755
17	40	0.6	0.10	FALSE	−39,450.0	589.735
18	40	0.6	0.15	FALSE	−38,800.0	892.371
19	50	0.2	0.05	TRUE	−37,100.0	1350.703
20	50	0.2	0.10	TRUE	−37,875.0	621.138
21	50	0.2	0.15	TRUE	−37,100.0	1194.809
22	50	0.4	0.05	FALSE	−38,525.0	945.103
23	50	0.4	0.10	FALSE	−38,675.0	713.041
24	50	0.4	0.15	FALSE	−38,100.0	500.202
25	50	0.6	0.05	FALSE	−38,575.0	933.385
26	50	0.6	0.10	FALSE	−38,200.0	1235.231
27	50	0.6	0.15	FALSE	−39,875.0	174.063

meets the criteria, the aircraft is assigned to a remote gate.

The best objective function value and thus a high efficiency solution for the problem, to achieve the assignment of aircraft to gates with reduced GHG emissions, is given by the following combination: {4, 8, 2, 5, 7, 1, 4, 0, 6, 5, 2, 8, 9, 1, 4, 7, 0, 5, 8, 2}. Moreover, this combination also provides the largest reduction in GHG emissions compared to the first-fit strategy assignment. Since the problem has three components for optimization, the best solution yields a minimum value for the three of them, and not just the one that leads to the greatest reduction in emissions. Additionally, both the first and second components indirectly contribute to reducing emissions. Indeed, by reducing the number of flights assigned to the apron area, emissions from buses transporting passengers from the apron area to the terminal are reduced. Moreover, by reducing the distance passengers need to travel to reach their connecting gate, the time to get there is also reduced, as are the chances of departure delays. The algorithm reduced CO₂ emissions by 1356.6 kg compared to unoptimized flight allocations, equivalent to an average fuel savings of about 19 liters per aircraft. For the calculation of CO₂ emissions, the Fuel Conversion Factor was used, which for the specific Jet-A fuel type is 3.16 Kg CO₂/Kg fuel (IATA, 2022).

Fig. 1 presents the results of the best GAP solution. On the y-axis of this chart are the gates and time on the x-axis is represented in minutes from 7:00 onwards. Each flight is represented by a rectangle whose length corresponds to the amount of time that the aircraft will spend at the gate until its next flight.

Flight-to-gate assignments have been created based on scheduled flight times, and only minor delays have been considered during the gate allocation process. In the case of significant delays, reassigning flights to new gates becomes necessary, even though rerunning the schedule with the updated arrival/departure times may disrupt the scheduled gate assignments and impact passengers' comfort. To minimize the impact of flight delays, the following strategy is adopted: the gate assignments for all flights will remain unchanged up to the flight directly preceding the delayed one; subsequently, any affected flights will be reassigned to

minimize deviations from the originally planned schedule, thereby reducing disruption. Considering that flight announcements on airport information boards are typically made 1–2 h before departure, the rescheduling process prioritizes minimal changes to the displayed schedule to ensure consistency and passenger convenience. For example, if flight No “TT8890” has a 1.5-hour delay (new arrival time Table 6), the updated gate assignment will be the following: [4, 8, 2, 5, 7, 1, 4, 0, 6, 5, 2, 8, 9, 2, 8, 7, 6, 5, 1, 0].

4.4. Sensitivity analysis

A sensitivity analysis was also performed to examine the impact of changing various factors on the optimization results (Fig. 2). Indicatively, based on the analysis, when the distance between gates is reduced to half, the optimized GHG emissions are reduced to 6521 (26 kg decrease) and the optimized walking distance is reduced to 5875 m (3875 m decrease). Also, when the distance between gates is doubled, the optimized GHG emissions are reduced to 5926 kg (621 kg decrease) and the optimized walking distance is equal to 0 (9750 decrease). Based on the results, it is observed that, as the distance between gates is reduced, the importance of this criterion in the objective function is less significant. When the airport becomes significantly larger and the distance between gates increases, the walking distance equals zero, as it makes sense to keep passengers from transit flights at the same gates.

In terms of GHG emissions, there is no standard trend. These insights are valid for this case where the number of transfer passengers is limited (65 pax). Regarding the sensitivity analysis of the distances from the runway to the gate and vice versa, it is normal to observe that as the distances traveled by airplanes increase, CO₂ emissions increase. The optimized taxiing distance, which refers to the sum of the distances of all aircraft from the runway to the gate after landing and vice versa from the gate to the runway before takeoff, also increases. It should be noted that this increase (or decrease) is not proportional because the objective function has multiple components. Also, the results of the sensitivity

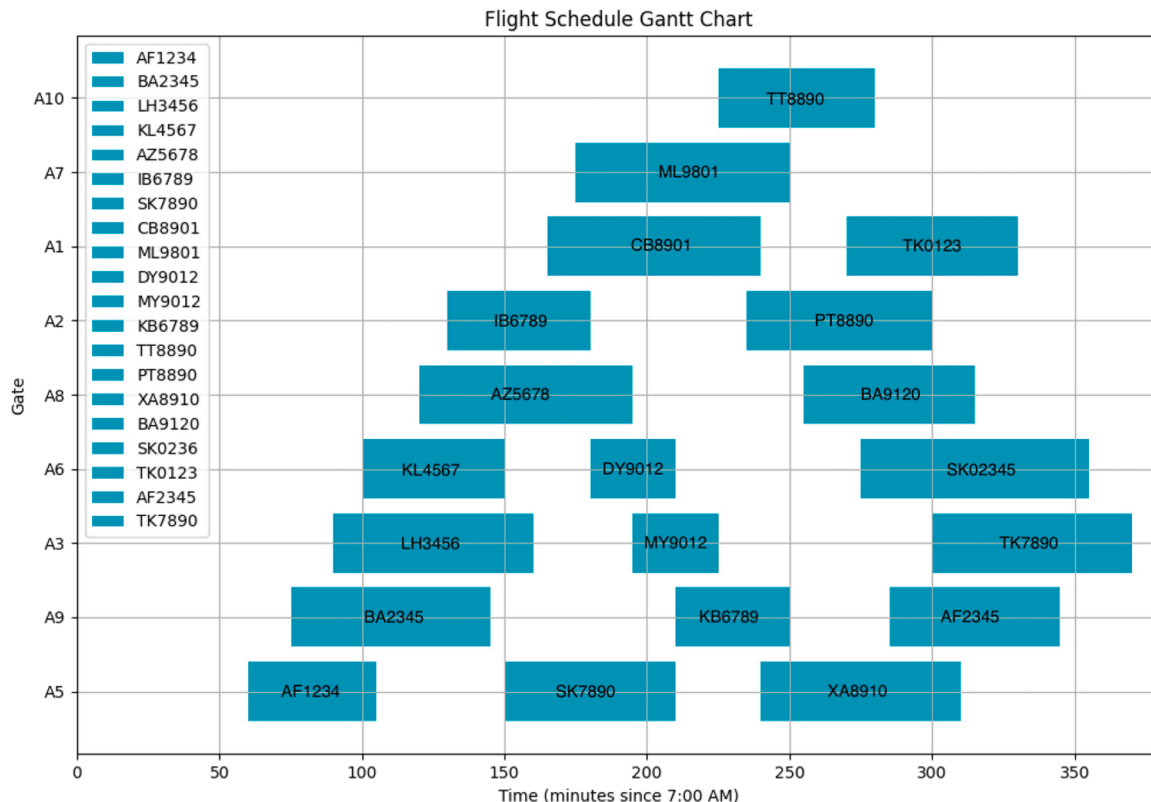


Fig. 1. Proposed GAP solution.

Table 6
Updated flight data after delay.

Flight ID	Arrival Time	Arrival Time (update)	Departure Time	Departure Time (update)	Aircraft Class	Destination	Connection	Fuel Consumption (kg/s)
TT8890	10:45	12:15	11:40	13:45	IV	Domestic	–	0208

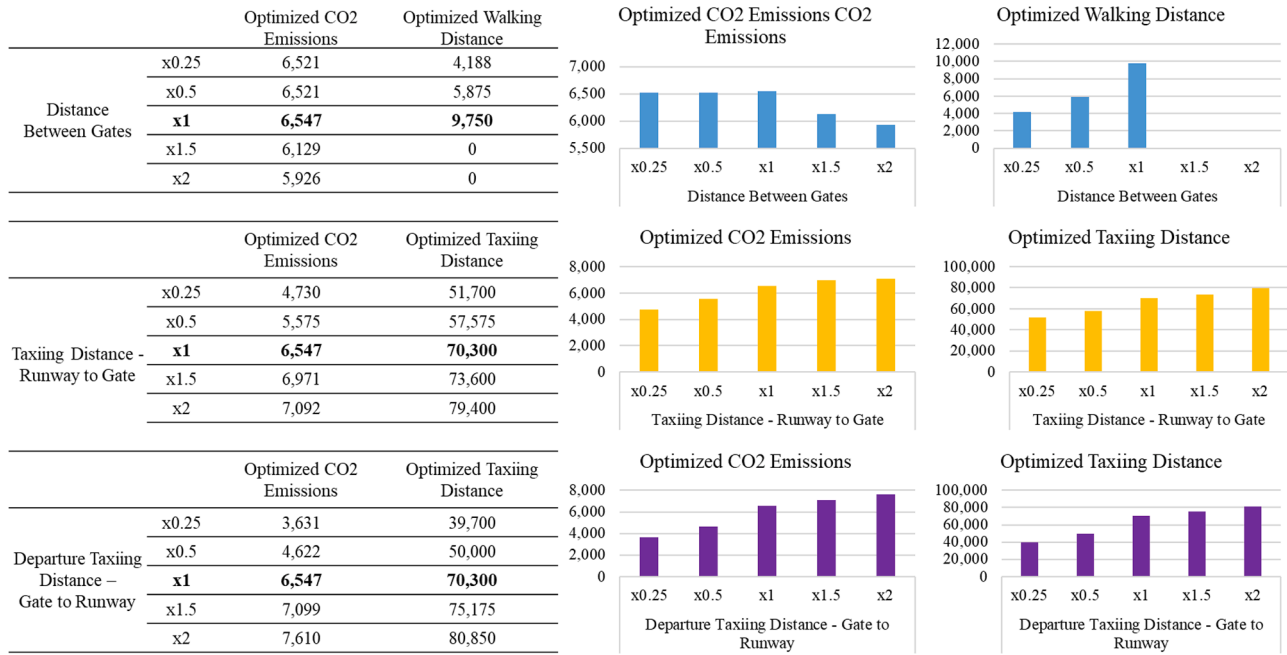


Fig. 2. Sensitivity analysis summary.

analysis indicate the need to meticulously plan airport layouts to achieve the best combination of optimized GHG emissions and passenger convenience. In terms of the sensitivity analysis regarding aircraft sizes, the results indicate that variations do not significantly affect optimized CO₂ emissions and optimized taxiing distances. Similarly, the sensitivity analysis on airport gates shows that optimized CO₂ emissions and operating taxiing distances are not affected by gate size. Fig. 2 shows a summary of the sensitivity analysis.

To further analyze the impact of integrating GHG emissions in the objective function, a comparative analysis has been provided between two optimization scenarios: (1) the full model, which includes GHG emissions in the objective function, and (2) a simplified version where emissions are excluded (i.e., $w_3 = 0$). The results indicate that in the second model, when emissions are not considered, aircraft are assigned to gates in a manner that reduces total passenger walking distances, by approx. 17 %. However, this comes at a cost since the taxiing distances and CO₂ emissions increase by approx. 3 %. This trade-off suggests that prioritizing emissions reduction modifies gate assignments in a way that unnecessary aircraft movements are reduced. In that context the optimization leans towards passenger convenience by reducing passenger walking distances. It highlights the interplay between environmental sustainability and passenger experience, emphasizing the need for a balanced approach in airport gate assignment strategies.

5. Conclusions

The assignment of airplanes to airport gates has a clear impact on operational costs, passenger satisfaction, and environmental sustainability. This paper proposed a model for tackling the associated gate assignment problem (GAP), which minimizes passenger walking efforts and GHG emissions; the model attempts to improve the carbon footprint of an airport and the effort required by travelers to access their connecting flight and demonstrates the significant impact of airfield design

on GHG emissions. A genetic algorithm with a Nash equilibrium control was exploited to efficiently and quickly solve the model. The model was demonstrated for a hypothetical airport accommodating a realistic flight schedule, with results demonstrating its performance against a non-optimized airplane-to-gate assignment. In addition, this paper combines passenger comfort with responsiveness by reducing the gate distance between disembarking from the first flight and boarding the second flight.

By modeling each flight as a self-interested agent seeking optimal gate assignment, airport managers can ensure that the final solution is stable because no single flight can unilaterally change gates to improve the overall performance. This approach enhances operational efficiency, reduces emissions primary with minimizing total distance, and minimizes gate conflicts, all while ensuring scalable and computationally tractable solutions. Importantly, it enables decision-makers to anticipate and avoid inefficiencies that could arise from suboptimal or unstable gate reassignments, particularly under growing traffic and tighter sustainability goals. With this game-theoretic perspective we want to introduce a new way of balancing competing objectives like passenger convenience, environmental impact, and operational constraints in real-world airport environments and open a new way for solve airport oriented problems into a more game theoretic framework.

It should be noted, however, that the model has certain limitations; additional factors and operational constraints encountered in the literature could have been included in the model, especially those targeting minimizing operational costs and/or maximizing revenues. Also, airline policies, such as low-cost airlines or gate leasing, and competition between airlines (Sun et al., 2024) are not considered in the model. Future research could, in that sense, try to balance the contradicting priorities of airports, airline companies, and passengers, along with sustainability aspects in the GAP formulation, and attempt to describe in greater detail passenger and aircraft ground movements affecting those factors.

CRediT authorship contribution statement

Stavroula Bagioneta: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Konstantia Kontodimou:** Writing – original draft, Methodology, Conceptualization. **Konstantinos Kepaptsoglou:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

No funding was received.

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